Dissertation Plan: Interactive Proofs for Resource-Constrained Quantum Learning

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1 Overview

This document outlines the planned structure of the final report for the CS310 project on interactive proofs for verifying quantum machine learning. The project investigates the theoretical and practical development of verification protocols that allow a classical client to check that an untrusted quantum server performed a correct quantum learning task, with a focus on the Mixture of Superpositions (MoS) technique introduced by Caro et al. (ITCS 2024).

Target length: 8,000 words (max 10,000). Front matter and back matter excluded from count.
Core thesis: The MoS verification protocol for agnostic parity learning can be faithfully simulated and scaled to n=16+ via Goldreich–Levin extraction, and its soundness and efficiency properties can be empirically characterised under realistic failure modes—including adversarial provers, noise, Fourier-dense label distributions, and increasing system size.

2 Chapter Plan

Word counts are approximate targets. The total for main-body chapters sums to approximately 8,000 words.

Chapter	Words	Content and Notes
Front Matter	N/A	**Title page** (title, author, supervisor, year). **Abstract** ($\leq$300 words): research question, approach, key results. **Keywords** (6–8): quantum machine learning, interactive proofs, Fourier sampling, MoS, classical verification, agnostic learning, Qiskit simulation. **Table of Contents.**
1. Introduction	~1,000	**1.1 Motivation and Context.** QML promises and the cloud-delegation trust problem. Why verification matters as quantum hardware scales. **1.2 Research Question.** Can a classical client verify that an untrusted quantum server performed the correct learning task? Focus on agnostic parity learning via MoS. **1.3 Objectives and Contributions.** (i) Implementation of MoS-based QFS in Qiskit; (ii) simulation of the Caro et al. verification protocol; (iii) empirical study of soundness under dishonest provers; (iv) scaling the protocol to $n \geq 10$ via Goldreich–Levin extraction; (v) efficiency analysis under Fourier-sparse vs. Fourier-dense regimes. **1.4 Report Structure.** Roadmap of chapters.

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Chapter	Words	Content and Notes
2. Background	∼1,800	**2.1 Quantum Computing Preliminaries.** Qubit model, measurement, Hadamard transform. Only material needed for the protocol. **2.2 Agnostic Learning and Fourier Analysis.** Boolean Fourier analysis, agnostic PAC model, parity learning (LPN hardness). Define α-agnostic learner, benchmark class $\mathcal{B}$. **2.3 Quantum Fourier Sampling.** Superposition examples, Hadamard + measure → approximate sampling from $\lvert\hat{\varphi}(s)\rvert^2$. Theorem statement (Thm. 4.1). **2.4 The Verification Problem.** QPIP model, completeness/soundness, why BQP verification is hard (oracle separations, $\mathrm{BQP}^A \not\subset \mathrm{MA}^A$). Computational vs. information-theoretic soundness. **2.5 Mahadev's Protocol (Brief Overview).** LWE, TCF functions, single-qubit commitment and Hadamard round. State the result without full proof. Explain why direct composition with MoS fails (mixed-state obstacle, multi-qubit OTP). Position as context, not a contribution of this project. **2.6 Related Work.** Caro et al. ITCS 2024, Ma–Su–Deng noise robustness, Caro et al. impossibility results (arXiv:2410.23969), compiled nonlocal games. Brief survey.
3. The MoS Verification Protocol	∼1,200	**3.1 Mixture of Superpositions Construction.** Definition of $\mathcal{F}_D$, the mixed state $\rho_D = \mathbb{E}_{f\sim\mathcal{F}_D}\left[\lvert\psi_f\rangle\langle\psi_f\rvert\right]$. Why MoS resolves the distributional vs. functional gap. **3.2 Approximate Quantum Fourier Sampling from MoS.** $H^{\otimes(n+1)}$ + measure → post-select on b=1. Theorem: output distribution $\approx \lvert\hat{\varphi}(s)\rvert^2$. Sample complexity $O(1/\vartheta)$. **3.3 The Three-Step Protocol.** Step 1: Prover extracts heavy coefficients L via QFS. Step 2: Verifier estimates correlations using classical examples. Step 3: Weight check and hypothesis construction. Information-theoretic soundness argument. **3.4 Protocol Parameters.** Relationship between ε, δ, ϑ, $\lvert L\rvert$, sample complexities. How these translate to experimental parameters.

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Chapter	Words	Content and Notes
4. Implementation	∼1,200	**4.1 Architecture Overview.** Module structure: `mos/` (simulator), `ql/` (prover + verifier). Design decisions: why Qiskit, why statevector + circuit + batched modes. **4.2 MoS Simulator (`mos_simulator.py`).** State preparation via oracle circuits ($H^{\otimes n} \otimes I$ then U_f). Per-shot f-sampling to realise the mixture. Noise injection model: $\varphi_{\text{eff}} = (1-2\eta)\varphi + \eta$. Three execution modes and their tradeoffs. **4.3 Quantum Prover (`prover.py`).** Post-selection on label qubit = 1. Direct thresholding extraction (for $n \leq 16$). Goldreich–Levin iterative bisection (for large n): bucket weight estimation, pruning strategy, internal threshold selection. Noise floor estimation via bottom-quartile heuristic. Discussion of crossover point between methods. **4.4 Classical Verifier (`verifier.py`).** Classical example oracle. Fourier coefficient estimation via $\mathbb{E}[(1-2y)(-1)^{\langle s,x\rangle}]$. Weight interval check with tolerance. Hypothesis construction (`FourierHypothesis` class). **4.5 Validation.** Comparison of empirical $\Pr[s \mid b{=}1]$ against theoretical distribution at small n. Statevector vs. circuit mode agreement. Density matrix purity checks. At larger n (≥ 10), validate GL extraction against ground-truth heavy coefficients; report false positive/negative rates as a function of sample budget.

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Chapter	Words	Content and Notes
5. Experiments and Results	∼2,000	**5.1 Experimental Setup.** Two regimes: *small-n* (n=4, direct thresholding, noisy parity with target $s^* = 1011$, $\eta = 0.15$) and *scaled* (n=10–20, GL extraction, varied Fourier structure). Prover copies, verifier samples, seeds. **5.2 Honest Baseline.** Protocol at n=4: prover identifies $s^* = 1011$, verifier accepts, hypothesis achieves near-optimal risk. Agnostic bound met. Then repeat at n=10 and n=16 with GL extraction to confirm the protocol scales: report post-selection rate, GL tree depth, extraction accuracy, and wall-clock time. **5.3 Soundness: Dishonest Prover Experiments.** Exp 1 — Oracle mismatch (n=4): prover uses wrong target parity. Verifier rejects across 1-bit, 2-bit, and trivial mismatches. Demonstrates information-theoretic soundness. Exp 2 — Noisy prover (oracle noise): sweep η from 0 to 0.5. Track $(1-2\eta)^2$ attenuation of $\lvert\hat{\varphi}(s)\rvert^2$. Map the accept→reject transition. Connection to Ma–Su–Deng threshold $\eta \leq 1/(10\vartheta)$. Run at both n=4 (direct thresholding) and n=12 (GL) to compare noise sensitivity across extraction methods. Exp 3 — Gate-level noise: depolarising noise on H, X, CX gates via Qiskit `NoiseModel`. Beyond existing theory (different noise model). **5.4 Scaling and Efficiency.** Exp 4 — Scaling the honest protocol: run full protocol sweep over $n \in \{4, 6, 8, 10, 12, 14, 16\}$ with GL extraction. Report: GL tree nodes explored, post-selection rate, prover sample cost, verifier sample cost, and hypothesis quality as functions of n. Compare empirical scaling against theoretical $O(1/\vartheta)$ and $O(\text{poly}(n))$ predictions. Exp 5 — Fourier-dense functions: construct bent functions ($\lvert\hat{\varphi}(s)\rvert = 2^{-n/2}\ \forall s$) for worst-case exponential scaling of $\lvert L\rvert$. Use functions with inverse-polynomial ϑ for intermediate regime. Demonstrate the protocol's efficiency collapse as Fourier sparsity decreases. Exp 6 — Verifier truncation: at scaled n, prover sends full list L, verifier processes only a random subset. Measure completeness loss (honest rejected) and soundness loss (dishonest accepted) as a function of fraction processed. Exp 7 — Average-case performance: sample ground-truth functions from a distribution at varying n, run protocol, report average acceptance rate and hypothesis quality. Distributional PAC flavour. **5.5 Results Summary.** Tables and figures summarising all experiments. Key findings. Explicit comparison of small-n and scaled regimes.

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Chapter	Words	Content and Notes
6. Discussion	∼800	**6.1 Interpretation of Results.** How empirical findings relate to theoretical bounds. Are the bounds tight? Where does the protocol break down in practice vs. theory? Compare small-n (exact enumeration) and scaled (GL extraction) regimes: does the GL estimator introduce systematic bias that the theory does not account for? **6.2 Scaling Observations.** What actually limits n in practice: post-selection rate decay, GL estimation noise, or wall-clock time? How does the empirical sample complexity compare to the theoretical $O(1/\vartheta)$? Discuss the practical crossover between direct thresholding and GL. **6.3 Limitations.** Weight interval still computed from true φ (not a distribution-class promise). Gate noise model differs from theoretical bit-flip model. Statevector simulation means no true quantum resource accounting. **6.4 The Mahadev Connection.** Brief discussion of why direct composition fails (mixed-state obstacle, multi-qubit OTP). This is a theoretical boundary, not a limitation of this project.
7. Conclusions and Future Work	∼300	**7.1 Summary of Contributions.** Working simulation of MoS verification protocol from n=4 to n=16+. Empirical evidence for soundness, noise sensitivity, scaling behaviour, and efficiency limits. Characterisation of GL extraction reliability at scale. **7.2 Future Work.** Real quantum hardware execution. Ma–Su–Deng error rectification implementation. Exploring non-uniform marginals. Non-parity concept classes. Tighter efficiency bounds via average-case analysis.
Back Matter	N/A	**References.** Aharonov et al. 2017, Mahadev 2018, Regev 2005/2024, Caro et al. ITCS 2024, Caro et al. 2024 (impossibility), Ma–Su–Deng 2024, + implementation references (Qiskit, etc.). **Appendices** (if needed). Selected code listings (circuit construction, key algorithms). Extended experimental data tables. **Author's Self-Assessment** (optional). ∼100-word answers to the five self-assessment questions.

3 Planned Figures and Tables

3.1 Chapter 2 — Background

- **Fig. 2.1:** Schematic of the QPIP client–server model (classical messages, quantum prover).
- **Fig. 2.2:** Mahadev's protocol flow — TCF commitment, challenge rounds, correction formula (conceptual diagram, not a contribution of this project).

3.2 Chapter 3 — Protocol

- **Fig. 3.1:** The MoS construction: sampling $f \sim \mathcal{F}_D$, preparing $|\psi_f\rangle$, averaging to get ρ_D.
- **Fig. 3.2:** Three-step protocol flow diagram (Prover → list L → Verifier → accept/reject + hypothesis).
- **Table 3.1:** Protocol parameters and their theoretical relationships (ε, δ, ϑ, $|L|$, sample complexities).

3.3 Chapter 4 — Implementation

- **Fig. 4.1:** Module architecture diagram (`mos/`, `ql/`, relationships between classes).
- **Fig. 4.2:** Example Qiskit circuit for n=3 MoS Hadamard measurement (from simulator demo), drawn using `quantikz2`.
- **Fig. 4.3:** Empirical vs. theoretical $\Pr[s \mid b{=}1]$ validation plot.

3.4 Chapter 5 — Experiments

- **Table 5.1:** Oracle mismatch results — prover target vs. verifier decision for each mismatch configuration.
- **Fig. 5.1:** Noisy prover sweep — plot of $(1-2\eta)^2$ attenuation and accept/reject boundary as a function of η, shown for both n=4 (direct) and n=12 (GL).
- **Fig. 5.2:** Gate-level noise — decision outcomes as a function of depolarising error rate p.
- **Fig. 5.3:** Scaling sweep — prover sample cost, verifier sample cost, GL nodes explored, and hypothesis risk as functions of n (n=4 to 16).
- **Fig. 5.4:** Bent function scaling — list size $|L|$ and verifier sample cost vs. n (exponential regime).
- **Fig. 5.5:** Verifier truncation — completeness and soundness as a function of fraction of L processed, at multiple n.
- **Table 5.2:** Average-case protocol performance over randomly sampled ground-truth functions at varying n.
- **Table 5.3:** Hypothesis risk vs. best parity risk across all experimental conditions and system sizes.

4 Alignment with Marking Criteria

4.1 Technical Content

Use of references: Chapter 2 provides a thorough survey of Mahadev (2018), Caro et al. (ITCS 2024, arXiv 2024), Ma–Su–Deng (2024), and the impossibility landscape. All claims are grounded in cited results.

Problem solving and tools: The implementation uses Qiskit circuits with statevector, circuit, and batched execution modes. The Goldreich–Levin extraction is exercised at n=10–16, demonstrating practical engagement with scalable algorithmic techniques beyond toy problem sizes.

Effective problem analysis: The experimental design (Sections 5.3–5.4) systematically isolates failure modes: oracle mismatch, oracle noise, gate noise, Fourier density, and verifier truncation. The scaling sweep (Exp 4) directly tests whether theoretical asymptotic predictions hold at intermediate n.

Technical achievement: Working end-to-end simulation of a protocol from a 2024 research paper, scaled from n=4 to n=16+. Novel experiments on scaling, truncation, and average-case performance suggested by the paper's authors.

Critical discussion: Chapter 6 critically evaluates limitations (idealised weight interval, gap between noise models and theory) and includes a dedicated scaling analysis (6.2) comparing empirical costs against theoretical predictions. The Mahadev connection is discussed honestly as an open problem, not overclaimed.

4.2 Communication Skills

Structure: The report follows a clear arc: motivation → theory → implementation → experiments → discussion. Each chapter leads naturally to the next. Sign-posting via chapter introductions and the protocol flow diagram.

Presentation: Figures and tables break up dense material and make experimental results immediately accessible. The planned ∼8,000-word length is within the required range.

5 Honest Assessment: Weaknesses and Risks

This section identifies where the project is most vulnerable under the marking criteria and flags structural risks that need to be addressed during writing.

5.1 Is this enough work?

With the scaling experiments (5.4) included, the project moves beyond reproduction into genuine experimental investigation. Running the full protocol from n=4 to n=16+ with GL extraction, characterising the efficiency collapse under bent functions, and testing verifier truncation at scale are original empirical contributions that the source paper does not provide. The project is sufficient for a solid mark *provided the scaling experiments are completed and discussed critically.* The risk is no longer "not enough work" but rather "scaling hits a wall and results are thin" — which is manageable, because negative or partial results are still reportable if analysed honestly.

5.2 Specific weaknesses against marking criteria

Technical Content

Innovative design — moderate. The implementation is a simulation of an existing protocol. There is no new algorithm, no new proof technique, and no new protocol design. The architecture choices (statevector vs. circuit modes, noise injection model, GL extraction) are sensible engineering but not novel. The scaling experiments and verifier truncation analysis are original experimental contributions, but an assessor specifically looking for design novelty may still find the project closer to "thorough empirical study" than "innovative system." To mitigate this: frame the GL extraction at scale and the truncation experiment as original work that required non-trivial engineering decisions (bucket weight estimation strategy, pruning thresholds, crossover heuristics).

Problem solving — strengthened by scaling. The move to larger n will surface real technical challenges: GL estimation noise propagating through the bisection tree, post-selection rate decay making sample budgets impractical, and potential divergence between the noise floor heuristic and reality at scale. These are genuine problems to solve and discuss. The report should treat them as first-class content, not footnotes.

Critical discussion — still a risk if superficial. Chapter 6 is now ∼800 words with a dedicated scaling analysis section (6.2), which is better. But the Mahadev discussion (6.4) still risks being hand-wavy if it just says "direct composition fails because mixed states." It needs to be precise: *which specific step*

of Mahadev's protocol breaks, and *why* the MoS mixed state is fundamentally incompatible with the spot-checking mechanism. Similarly, the limitations section (6.3) must assess *how much* each limitation affects the conclusions, not just list them.

Communication Skills

Background length — risk of imbalance. Chapter 2 is allocated ∼1,800 words, which is 22% of the total budget. For a project whose contribution is primarily experimental, this is borderline too much. If the background reads as textbook exposition rather than targeted preparation for the experiments, assessors may see it as padding. Every paragraph in Chapter 2 should clearly serve a later chapter. For instance: the oracle separation material ($\mathrm{BQP}^A \not\subset \mathrm{MA}^A$) is interesting context but if it takes 300 words and is never referenced again, cut it.

Mahadev coverage — scope risk. Section 2.5 on Mahadev's protocol is a high-wire act. Too little and the assessor wonders why it is mentioned at all. Too much and it looks like the report is trying to claim territory the project does not actually occupy. The Mahadev material should occupy at most 400 words and should exist *only* to set up Section 6.4's discussion of why direct composition fails.

Code documentation. The marking criteria explicitly mention "software and hardware documentation." The current plan does not allocate space for this beyond Chapter 4. If the appendix includes code listings, they need clear inline comments and the report text must explain *design decisions*, not just describe what the code does. Why is the noise floor estimated from the bottom quartile rather than the median? Why does GL use an internal threshold of $\vartheta/2$? These are the kinds of questions assessors ask.

5.3 Remaining risk: what if scaling hits a wall?

The most likely failure mode is that GL extraction at $n \geq 14$ becomes unreliable due to estimation noise in bucket weights, or that the sample budget required for accurate post-selection becomes computationally impractical in statevector simulation. If this happens, the report should:

1. Report *where* the wall is (which n, which component fails first).
2. Analyse *why* it fails (estimation variance? post-selection rate? simulation time?).
3. Compare the observed wall against the theoretical predictions.

This is still a valid and interesting result. The worst outcome is not "scaling failed" but "scaling failed and the report does not discuss why."

5.4 What would make this a strong dissertation

The scaling work addresses the most critical gap. To push from solid into strong, the report needs at least one of:

1. **A genuine empirical finding.** For example: showing that the accept/reject transition under noise occurs at a different threshold than Ma–Su–Deng predict, or that verifier truncation to 50% of L preserves soundness in practice despite the theory offering no guarantee, or that GL extraction degrades gracefully rather than catastrophically as n increases.
2. **Tight integration of theory and experiment.** Every experiment should reference a specific theoretical claim and either confirm it, refine it, or show where it breaks. The report should read as a dialogue between the mathematics and the simulation, not as "here is the theory" followed by "here are some plots."